

• Audit: checking what happened in a system.

Compliance: checking if you are following rules / laws / security standards

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LOGS FUNDAMENTALS:

- Logs are the digital footprints left behind by any activity performed on a digital device, this activity can be normal one or a malicious one. Logs are the most important traces left behind after any incident.
- Use cases of logs:
 - a) Security events monitoring: Logs helps us detect anomalous behaviour when monitoring a real time events.
 - b) Incident Investigation and Forensics: Logs tells everything what happened during the incident, so they are used by investigation team for root cause analysis.
 - c) Troubleshooting: Logs also stores errors in system or application and can be used to diagnose issues and helpful in fixing them.
 - d) Performance monitoring: Logs can also be used to monitor the performance of any application.
 - e) ^{ti}Auditing and compliance: Logs helps us in auditing and compliance, because they make it easier with its capability to establish a trail of different kinds of activities.
- Types of logs:

Logs are classified into different types on the basis of the information they provide. Some of these logs are:

Contains logs of different running operations that are of OS itself.

- a) **System logs:** These logs are used to troubleshoot running issues of OS. Ex: Hardware events, System error events, Driver loading events, system startup and shutdown events.
 - b) **Security logs:** These logs provide information related to security of a system. Ex: Authentication and authorization events, Security policy changes events etc.
 - c) **Application logs:** It contains information about the specific events of the application such as interactive or non-interactive activity happening inside a application. Ex: Application update, changes and error events.
 - d) **Audit logs:** These logs provide detailed information on the system changes and user events. Ex: Data access events, system change event, user activity event etc.
 - e) **Network logs:** They provide information on the network's outgoing and incoming traffic. Ex: Incoming and outgoing traffic events, Network connection logs, network firewall logs.
 - f) **Access logs:** These logs provides detailed information about the access to different resources. Ex: Webserver access logs, Database access logs, Application and API access logs.
- **Windows event logs Analysis:**
Windows OS has a utility called "event viewer" which gives a nice graphical user interface to view and search anything in these logs.

The windows event log has different fields:

Description: This field has a detailed information of the activity.

Log name: It indicates the log file name

Logged: Time of activity

Event id: Event ID's are unique identifiers for a specific activity.

Some important Event ID's in windows OS are:

4624 : A user account successfully logged in

4625 : " " " failed to login

4634 : " " " successfully logged off

4724 : An attempt was made to reset account's password and many more.

Event viewer allows us to search for logs with a specific event ID with its 'Filter Current log' feature.

• Webserver access logs analysis:

When we visit a website we make different types of requests there. All these requests are logged by the website and stored in a log file on the webserver on which the website is running.

The log contains the following information:

IP address: Who made the request

Timestamp: The time when the request was made

Request: The request details

Status code: The response from the server

User-agent: user's OS and browser etc when making the request